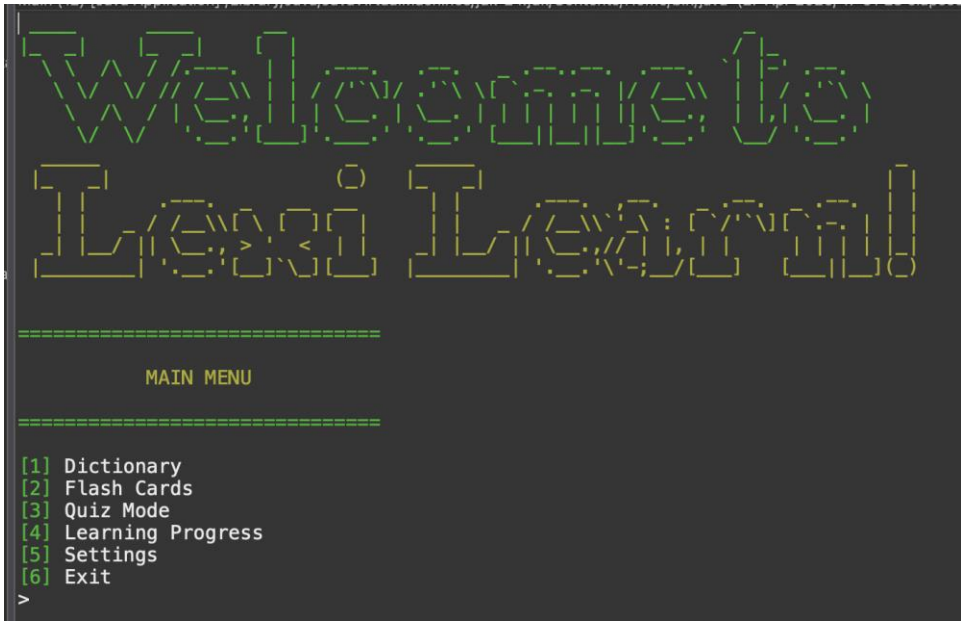
Lexi Learn

Lexi Learn is an English learning console app that is designed for learning English vocabulary using a free dictionary API.

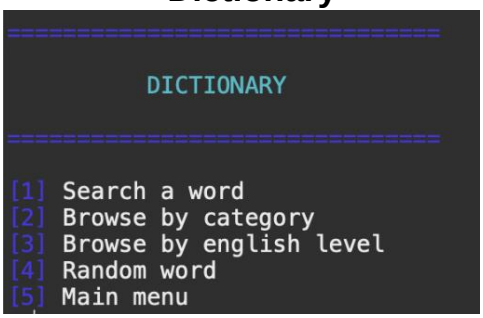
Base Url: https://api.dictionaryapi.dev/api/v2/entries/en/ + word to search

First, it greets the user with a welcome ASCII art, and the main menu

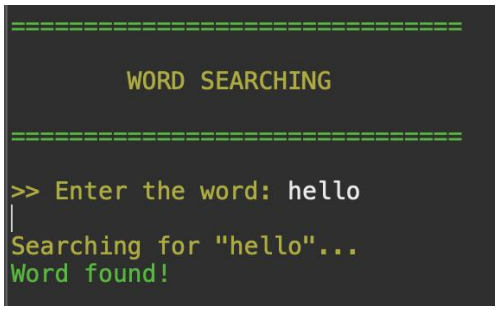


The app incorporates different word-searching options, such as browsing words by category, English level, free search, or a random word from the saved ones.

Dictionary



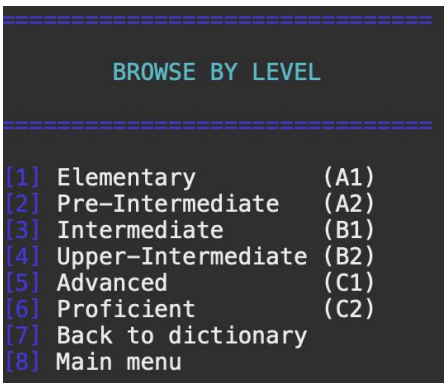
Free search

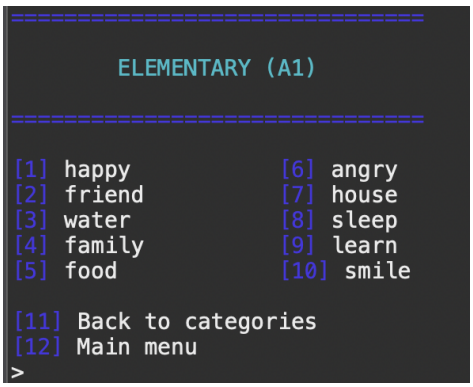


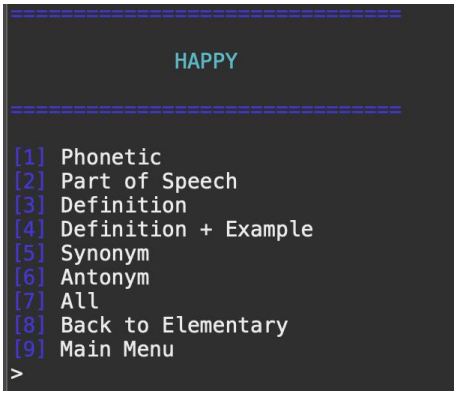
Random word



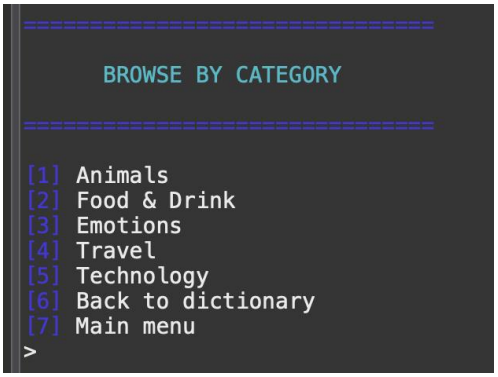
Browse by English level

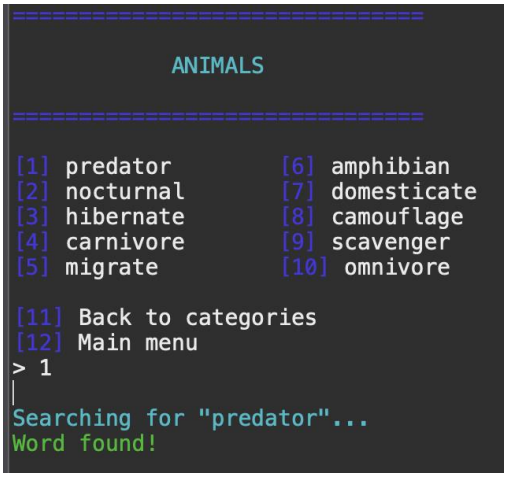


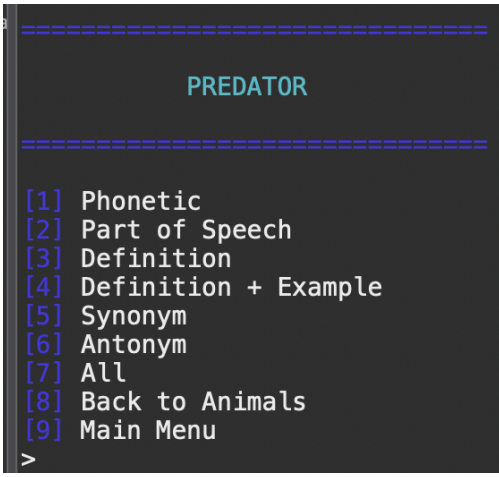




Browse by Category

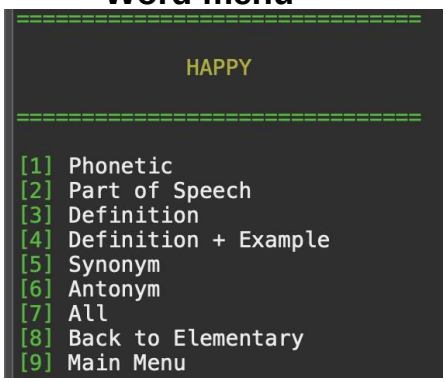




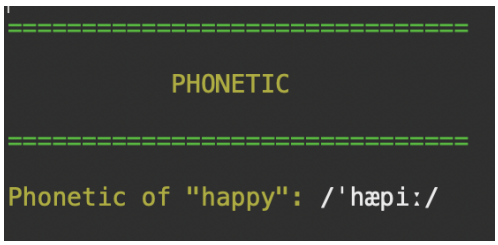


When the word is fetched from the API or a cache file, it shows the menu of word detail options, such as phonetic, part of speech, definition, definition + example, synonyms, antonym, and all (all data about the word fetched from the API).

Word menu



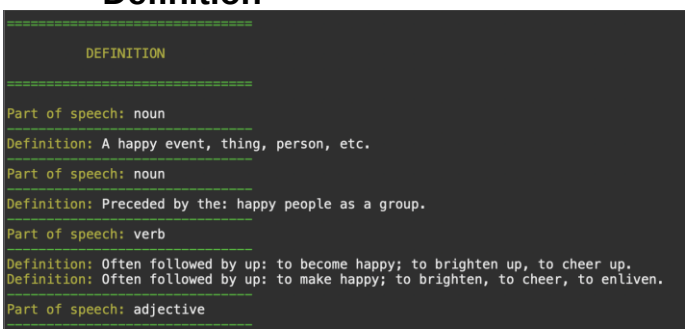
Phonetic



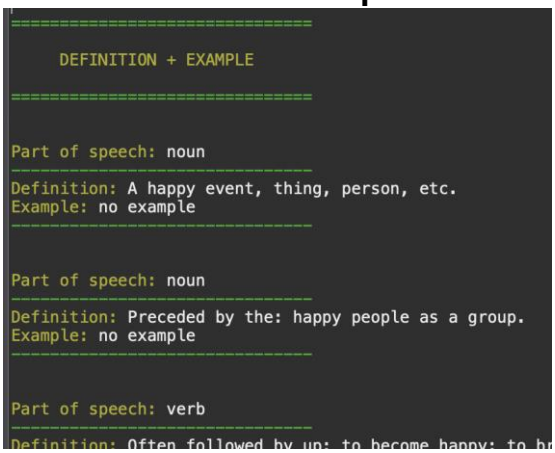
Synonyms



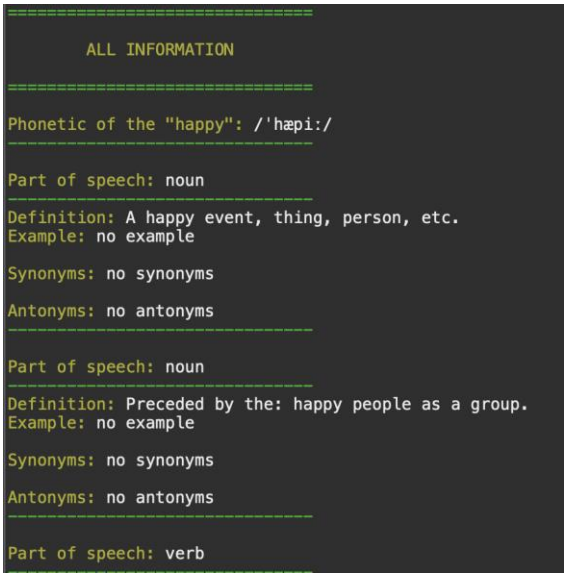
Definition



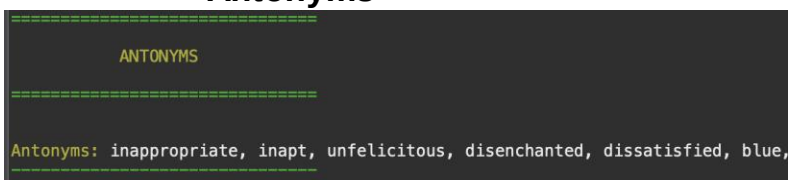
Definition + Example



All information

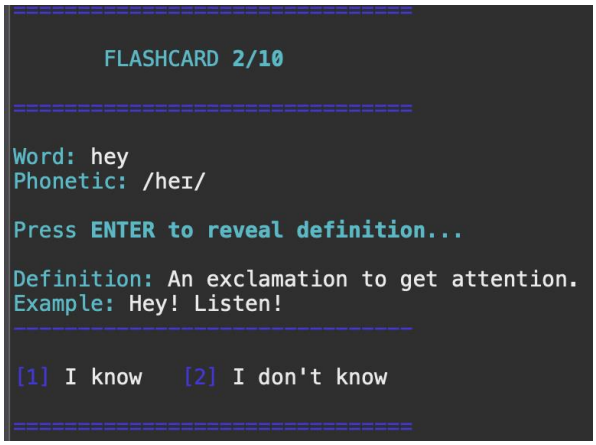


Antonyms

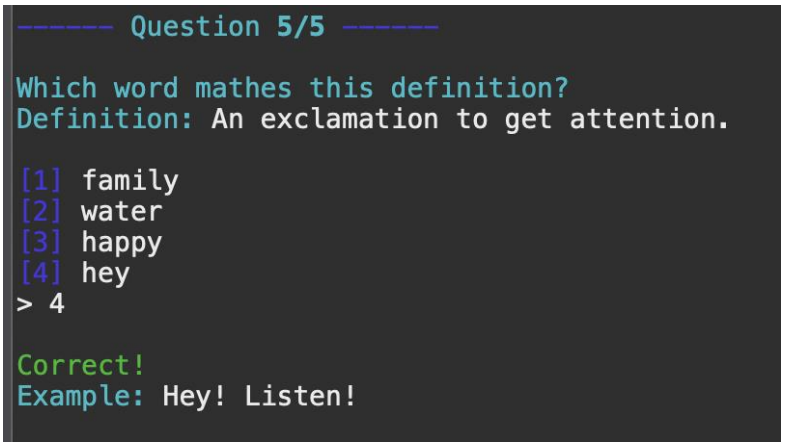


Besides the dictionary, it has two types of exercises to practice previously looked-up words: flashcards and quizzes. Flashcards allow you to test yourself and see how many words previously searched you have memorized. Quiz mode works based on the same saved words file as flashcards, but it tests your knowledge through multiple-choice questions, where you have to match the definition to the correct word.

Flashcards

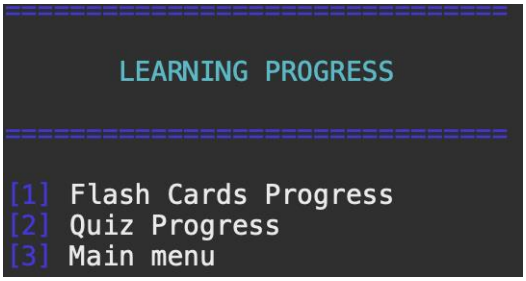


Quiz

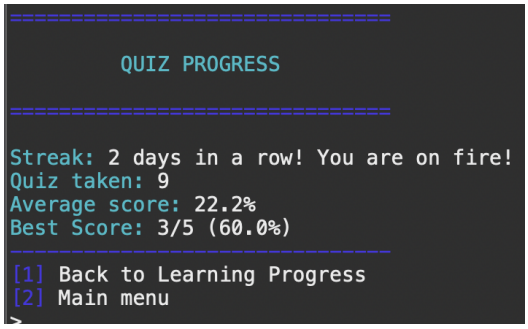


Moreover, Lexi Learn allows you to track your learning progress and the number of consecutive days you have been studying, based on both flashcards and quiz results.

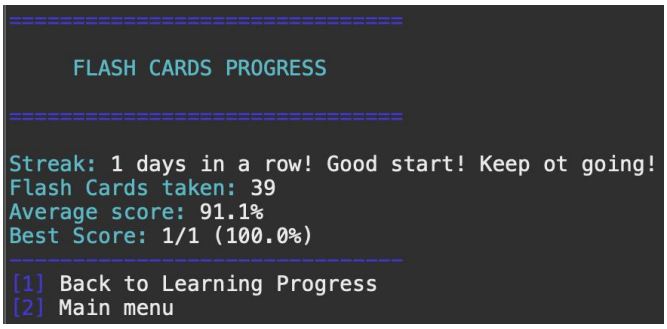
Learning Progress



Quiz Progress



Flashcards progress



Finally, the app has adjustable settings where you can set a color theme, control the difficulty and style for both flashcards and quizzes, the amount of information displayed when a word is searched, and have the option to reset all settings to their defaults.

Settings

```
==================================

          SETTINGS

==================================

[1] Color Theme
[2] Flash Cards Style
[3] Flash Cards Count
[4] Quiz Difficulty
[5] Word Detail
[6] Reset to Default
[7] Main menu
```

Color theme (forest)

```
==================================

          COLOR THEME

==================================

[1] Ocean (cyan + blue)
[2] Forest (green + yellow)
[3] Sunset (red + yellow)
[4] Default (white)

Current theme: Forest
----------------------------------
[5] Back to settings
[6] Main menu
```

Color theme (sunset)

```
==================================

          COLOR THEME

==================================

[1] Ocean (cyan + blue)
[2] Forest (green + yellow)
[3] Sunset (red + yellow)
[4] Default (white)

Current theme: Sunset
----------------------------------
[5] Back to settings
[6] Main menu
```

Flashcards style

```
==================================

        FLASH CARDS STYLE

==================================

[1] Short (words only)
[2] Basic (words + phonetic)
[3] Detailed (words + phonetic + part of speech

Current style: Basic
----------------------------------
[4] Back to settings
[5] Main menu
```

Flashcards count

```
==================================

        FLASH CARDS COUNT

==================================

[1] Quick (5 cards)
[2] Standard (10 cards)
[3] Deep dive (all saved words)

Current count: Standard
----------------------------------
[4] Back to settings
[5] Main menu
```

Quiz difficulty

```
==================================

        QUIZ DIFFICULTY

==================================

[1] Easy (5 questions, 4 options)
[2] Medium (10 questions, 5 options)
[3] Hard (all saved words, 6 options)

Current difficulty: Easy
----------------------------------
[4] Back to settings
[5] Main menu
```

Word detail

```
==================================

          WORD DETAIL

==================================

[1] Brief (1 entry)
[2] Standard (2 entries)
[3] Full (everything available)

Current word detail: Brief
----------------------------------
[4] Back to settings
[5] Main menu
```

Reset

```
Settings reset successfully!

==================================

          SETTINGS

==================================

[1] Color Theme
[2] Flash Cards Style
[3] Flash Cards Count
[4] Quiz Difficulty
[5] Word Detail
[6] Reset to Default
[7] Main menu
```

Lexis Learn saves searched words to a flashcards file for practice and to a cache file, from which words can be retrieved when the API is unreachable.

Fetching from cach file

```
==================================

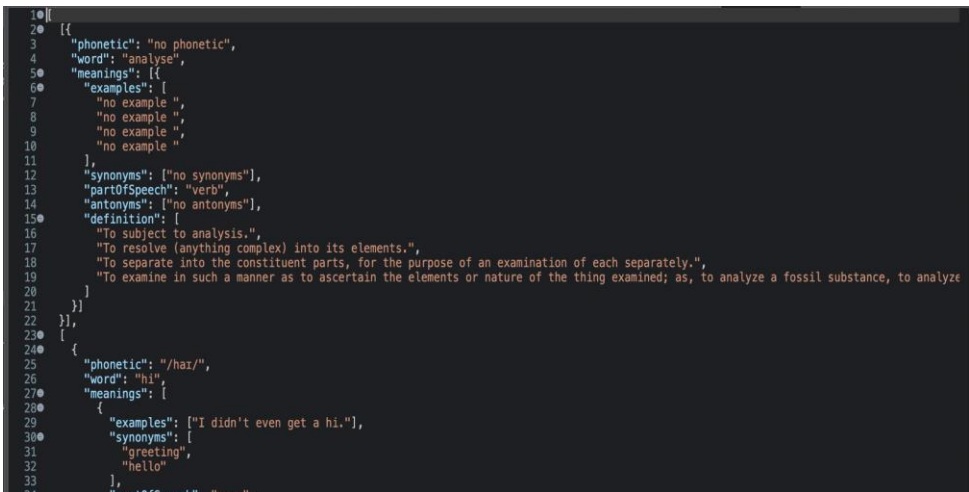
        WORD SEARCHING

==================================

>> Enter the word: hey

Searching for "hey"...
No internet connection!
Checking cache...
Searching for "hey"...
Loaded "hey" from cache
```

cache.json




```
[{
  "phonetic": "no phonetic",
  "word": "analyse",
  "meanings": [{
    "examples": [
      "no example ",
      "no example ",
      "no example ",
      "no example "
    ],
    "synonyms": ["no synonyms"],
    "partOfSpeech": "verb",
    "antonyms": ["no antonyms"],
    "definition": [
      "To subject to analysis.",
      "To resolve (anything complex) into its elements.",
      "To separate into the constituent parts, for the purpose of an examination of each separately.",
      "To examine in such a manner as to ascertain the elements or nature of the thing examined; as, to analyze a fossil substance, to analyze
    ]
  }]
}],
[
  {
    "phonetic": "/haɪ/",
    "word": "hi",
    "meanings": [
      {
        "examples": ["I didn't even get a hi."],
        "synonyms": [
          "greeting",
          "hello"
        ],
```

It also uses several others json files for storing settings, progress, category, and English level words.

category.json



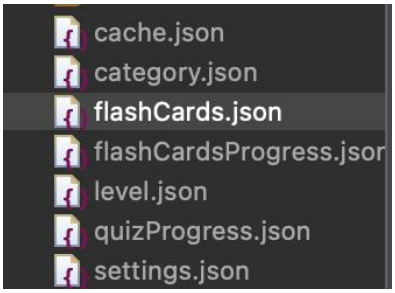
flashCardsProgress.json



settting.json

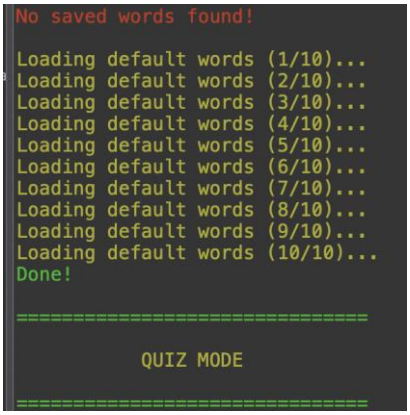


json files used

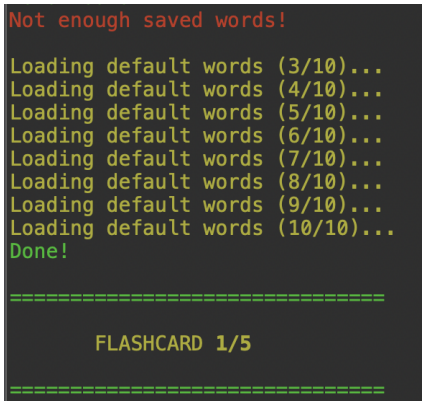


On this project, I really focused on making the app run consistently and not crash when an unexpected event occurs. To achieve this, I implemented various checks, including input validation, API response handling, file existence verification, empty file handling, offline mode support, and it even handles the case when the user does not have enough saved words to start a practice session, the automatically fills the flashcards file with the words from bult-in category or level word lists.

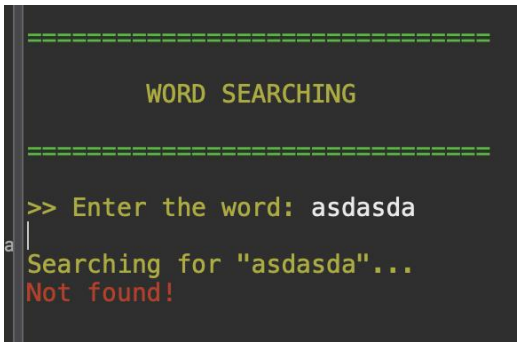
flashCards.json is empty



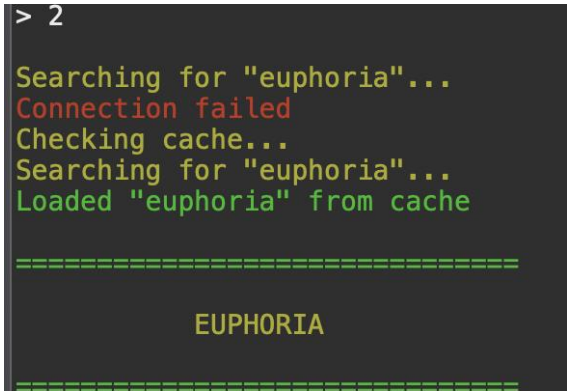
file has not enough words



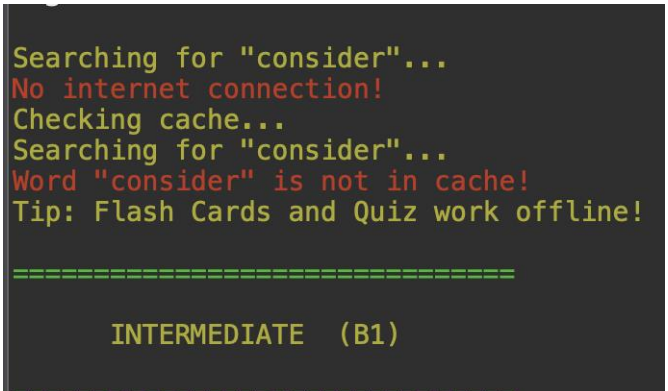
invalid



API fails, loading from cache



API fails, and not in cache, suggest a tip



When you exit Lexi Learn, it displays a goodbye message using ASCII art

